MIPS Assembly Homework I

Write MIPS assembly programs using the MARS similator to calculate the expressions below. For each expression, you must provide four versions:

I. **Immediate & Basic Instructions Version:** Set registers to immediate values first and then calculate upon registers. Use only basic (physical) instructions.
II. **Immediate & Pseudo-Instructions Version:** Set registers to immediate values first and then calculate upon registers. Utilize pseudo-instructions.
III. **Memory & Basic Instructions Version:** Store input numbers as variables in the .data segment, load them into registers, calculate, and store the result back into a labeled memory location. Use only basic (physical) instructions.
IV. **Memory & Pseudo-Instructions Version:** Store input numbers as variables in the .data segment, load them into registers, calculate, and store the result back into a labeled memory location. Utilize pseudo-instructions.

Arithmetic Expressions

a) 12 + 22 + 32 + 42

b) 150 - 60 - 45 - 15

c) (15 + 25) - (9 + 4 - 2)

d) (-30) + (-10) - ((-8) - (-12) + 25)

e) (-15) - (60 - (20 - (45 - 5)))

Bitwise & Logical Expressions

f) (45 & 25) ^ (~10 | 2)

g) ~(18 | 12) & (22 | 8 | ~(~14 & 6))

h) (0x5A & 0xFF) | ((0xC3 & 0x0F) ^ 0x0B)

Note: Use and (&), or (|), xor (^), and nor (for NOT (~) operations if you use basic instructions.)